

1 HSR-B/C/D 台車の加速度制約

HSR-B/C/D 台車のヤコビアンは次の通り.

$$\begin{pmatrix} \dot{x}_c \\ \dot{y}_c \\ \dot{\phi}_c \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} & 0 \\ J_{21} & J_{22} & 0 \\ \frac{r}{W} & -\frac{r}{W} & -1 \end{pmatrix} \begin{pmatrix} \omega_R \\ \omega_L \\ \omega_S \end{pmatrix} \quad (1)$$

$$\begin{aligned} J_{11} &= \frac{r \cos \theta_v}{2} - \frac{rs \sin \theta_v}{W} \\ J_{12} &= \frac{r \cos \theta_v}{2} + \frac{rs \sin \theta_v}{W} \\ J_{21} &= \frac{r \sin \theta_v}{2} + \frac{rs \cos \theta_v}{W} \\ J_{22} &= \frac{r \sin \theta_v}{2} - \frac{rs \cos \theta_v}{W} \end{aligned} \quad (2)$$

更に台車座標系での加速度を考える.

$$\begin{pmatrix} \ddot{x}_c \\ \ddot{y}_c \\ \ddot{\phi}_c \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} & 0 \\ J_{21} & J_{22} & 0 \\ \frac{r}{W} & -\frac{r}{W} & -1 \end{pmatrix} \begin{pmatrix} \alpha_R \\ \alpha_L \\ \alpha_S \end{pmatrix} + \begin{pmatrix} \dot{J}_{11} & \dot{J}_{12} & 0 \\ \dot{J}_{21} & \dot{J}_{22} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \omega_R \\ \omega_L \\ \omega_S \end{pmatrix} \quad (3)$$

$$\begin{aligned} \dot{J}_{11} &= \omega_S \left(-\frac{r \sin \theta_v}{2} - \frac{rs \cos \theta_v}{W} \right) \\ \dot{J}_{12} &= \omega_S \left(-\frac{r \sin \theta_v}{2} + \frac{rs \cos \theta_v}{W} \right) \\ \dot{J}_{21} &= \omega_S \left(\frac{r \cos \theta_v}{2} - \frac{rs \sin \theta_v}{W} \right) \\ \dot{J}_{22} &= \omega_S \left(\frac{r \cos \theta_v}{2} + \frac{rs \sin \theta_v}{W} \right) \end{aligned} \quad (4)$$